

THE STORY OF GEARS

SIMPLE BUT EFFECTIVE MECHANISMS FOR TRANSMITTING FORCES OF ROTATION, GEARS HAVE A LONG AND COMPLEX HERITAGE

The ability to alter the direction of a force, transmit it from one axis of rotation to another, or trade force with movement, is a vital aspect of many modern machines. Yet, such mechanical functions often rely on gearing techniques that are centuries old.

A gear is a wheel mounted on a central rotating axis, with a series of teeth or cogs around its outer edge that can engage with the cogs of another gear. The teeth allow the gear to transmit its angular motion to its neighbor, forming a pairing known as a transmission. The ratio of teeth between the two wheels determines the speed and force with which the second gear rotates, providing a so-called “mechanical advantage”; a smaller secondary gear rotates more rapidly, but with less torque or rotational force.

The earliest specific devices known to have used gears were the Chinese south-pointing chariots—direction-finding devices used in the 3rd century BCE.

Leonardo da Vinci's vehicle

This model of self-propelled automobile was built from a Leonardo da Vinci sketch. The force from the expansion of two wound springs is transmitted through ingenious gearing to drive the rear wheels.

1480

Leonardo da Vinci's work on gears

Italian polymath Leonardo da Vinci utilizes complex gear assemblies in many of his inventions, such as lens-grinding and metal-rolling devices, and shows a deep theoretical understanding of their function.

18th century Industrial Revolution

The rise of steam power during the Industrial Revolution drives advances in gear technology. The linear motion of steam pistons is applied to the rotation of locomotive wheels.

Steam locomotive



In ancient Greece, gear technology reached its apex with devices such as the Antikythera mechanism—a complex astronomical calculator recovered from a Mediterranean shipwreck around 1900.

PRACTICAL APPLICATIONS

More immediate practical applications of gears, such as their use in harnessing power from flowing water and wind, gradually spread throughout the ancient world. Treadmills powered by animals—or even humans—became common. Gears found various applications in mills—flour mills are perhaps the most familiar, but sawmills also used gears to turn rotating cutting blades, and hammer mills used gearing to lift and drop heavy hammers for beating metal or minting coins.

New advances led to the development of traditional clockwork in Europe in around the 13th century, and the Industrial Revolution saw the development of further ingenious transmissions to harness the power of steam engines. The use of gears has continued to the present day in modern machines ranging from automobiles to inkjet printers.

1781

Murdoch's gearing

Scottish engineer William Murdoch's sun and planet gear transmission converts vertical motion, such as that of a steam-driven beam, into the rotational motion of a driveshaft.

Sun and planet gearing



1835

Gear hobbing process

British engineer Joseph Whitworth invents hobbing—the first process for the production of high precision gears, on an industrial scale.



Plastic gear

1990s

Nanotechnology

Machines created on a nanoscale often rely on the same gearing principles as larger devices, except that the gear wheels are mere micrometers across.

1950s

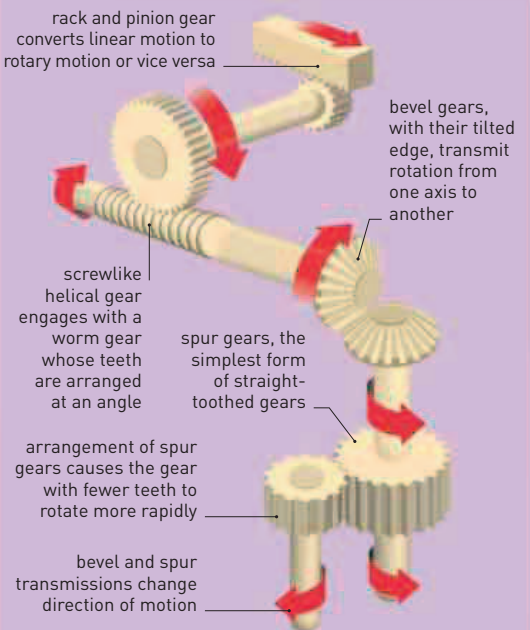
Plastic gears

Gears made from new plastics materials are introduced from the 1950s. They lack the strength of properly machined metal gears, but are far more easily and cheaply manufactured.

"Safety" bicycle



rack and pinion gear converts linear motion to rotary motion or vice versa



bevel gears, with their tilted edge, transmit rotation from one axis to another

screwlike helical gear engages with a worm gear whose teeth are arranged at an angle

spur gears, the simplest form of straight-toothed gears

arrangement of spur gears causes the gear with fewer teeth to rotate more rapidly

bevel and spur transmissions change direction of motion

TYPES OF GEARS

Gears can be designed or arranged in a variety of ways to transmit motion from one direction to another. Complex transmission assemblies can take the motive power from a single rotating drive shaft and apply it to drive a range of linear movements or to propel further rotating shafts at any speed required.